Tutorial 1

Contour Integration

1. Evaluate the integral $\int_C \left(z^2 + 3z\right) dz$ along the following contours C in the complex plane.
   (a) C is the straight line joining 2 to $2i$.
   (b) C is consists of the straight line from 2 to $2 + 2i$ and then to $2i$.
   (c) C is the quarter circle $|z| = 2$ from 2 to $2i$.

2. Evaluate the following integrals from the definition, i.e parametrise the curve C in terms of t and then calculate $\int_C f(z)dz = \int_a^b f(z(t))\frac{dz}{dt}dt$
   (a) $\oint_C z^2 d =$ where C is the circle centred at the origin, radius 2, starting and ending at 2.
   (b) $\oint_C \frac{1}{z} dz$ where C is
      (i) the square with corners $\pm 1, \pm i$
      (ii) the square with corners $\pm 1 \pm i$
      (iii) the unit circle
   (c) $\oint_C \ln z dz$ where C is the unit circle starting and ending at -1 .
   (d) Find $\oint_C f(z)dz$ where $f(z) = \pi e^{\pi \bar{z}}$ and C is the boundary of the square with verticies at the points $0, 1, 1 + i, i$ and the orientation in a counterclockwise direction. Discuss the analyticity of $f(z)$.

Recommended Additional Problems:

Zill & Shanahan: Ex 5.2, p.254-255, Q 1-24 (odd numbers).